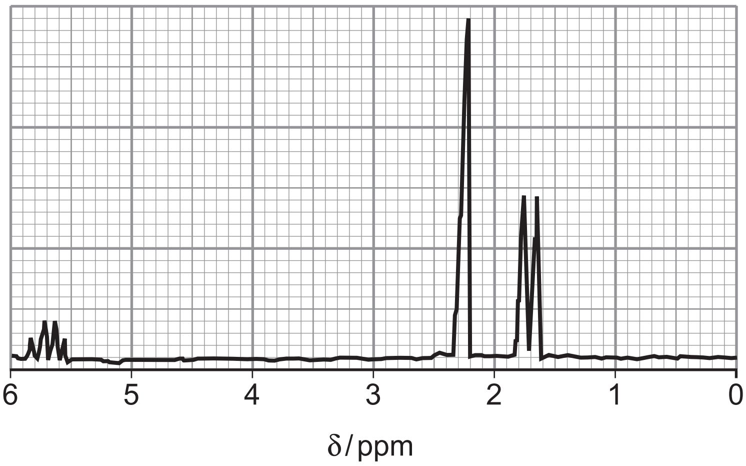
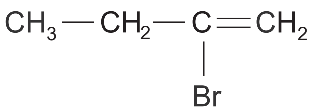
Examiner only

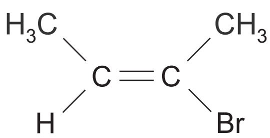
13. *(a)* The analysis of a compound shows that its formula is C_4H_7Br.
The high resolution ^1H NMR spectrum of the compound is shown below.



(i) A student suggested that the compound might be one of the following.

compound **L** $BrCH_2—CH_2—CH{=}CH_2$

compound **M** 

compound **N** 

Discuss the splitting pattern seen in the ^1H NMR spectrum to decide whether the compound is **L**, **M** or **N**. Give reasons for your answer. [4]

....

....

....

....

....

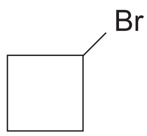
....

....

....

Examiner only

(ii) Another student suggested that the compound was bromocyclobutane.



I. Describe a chemical test that would show that the compound was either **L**, **M** or **N** and not bromocyclobutane. You should state the result of your test with each compound. [1]

....

....

....

II. Explain how the ^{13}C NMR spectrum would show that the compound was **L**, **M** or **N** and not bromocyclobutane. [2]

....

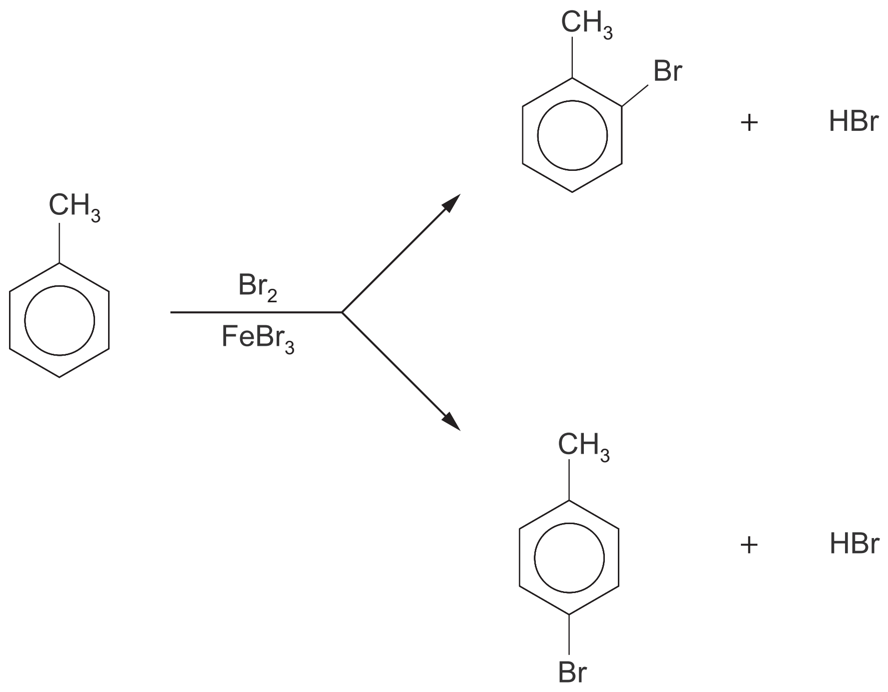
....

....

....

Examiner only

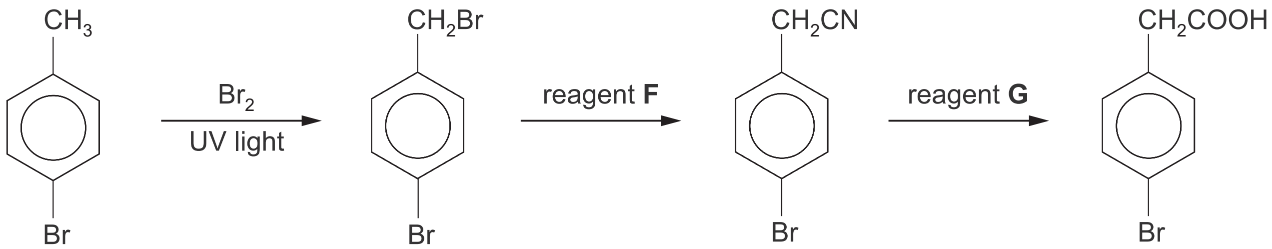
(b) The bromination of methylbenzene using electrophilic substitution gives a mixture of 2-bromomethylbenzene and 4-bromomethylbenzene.



Give the mechanism of the reaction that produces 4-bromomethylbenzene. [3]

Examiner only

(c) A reaction sequence to produce 4-bromophenylethanoic acid is shown below.



(i) The bromination of 4-bromomethylbenzene is a free radical process.

Give the equation for this reaction and then use the equation to calculate the minimum mass of bromine needed to convert 0.15 mol of 4-bromomethylbenzene to 4-bromo-1-(bromomethyl)benzene in the first stage of this reaction sequence. [2]

Mass of bromine = ... g

(ii) State the name of reagent **F**. [1]

....

(iii) State the name of reagent **G**. [1]

....

14

END OF PAPER